

1 Solution — GIAM 8.1

Problem 1

1.1 Problem

Name four sets in the equivalence class of $\{1, 2, 3\}$.

1.2 Setup — What is the Equivalence Class?

Section 8.1 defines two sets to be *equivalent*, written $A \equiv B$, iff there is a bijection $f : A \rightarrow B$. For *finite* sets this is the same as having the same number of elements: any bijection pairs the elements of A off with the elements of B one-to-one, so $|A| = |B|$; conversely, if $|A| = |B| = k$ we can list the elements as $A = \{a_1, \dots, a_k\}$ and $B = \{b_1, \dots, b_k\}$ and define $f(a_i) = b_i$.

The *equivalence class* of $\{1, 2, 3\}$ — written $\overline{\{1, 2, 3\}}$ or $\{1, 2, 3\}/\equiv$ — is

$$\overline{\{1, 2, 3\}} = \{S : S \equiv \{1, 2, 3\}\} = \{S : |S| = 3\}.$$

In words: it is the collection of *all* 3-element sets. Naming four members just means naming four three-element sets.

1.3 Four Sets in the Class

Any of the following work (and many more — see the remark below):

$$\{a, b, c\}, \quad \{\square, \triangle, \circ\}, \quad \{\text{red, green, blue}\}, \quad \{\pi, e, \sqrt{2}\}.$$

Each has exactly three elements, hence each is in bijection with $\{1, 2, 3\}$ via the obvious “list-and-match” pairing.

1.4 Bijections to $\{1, 2, 3\}$

To make explicit why each set is in the equivalence class, here is an exhibited bijection from $\{1, 2, 3\}$ to each one:

Set S	Bijection $f : \{1, 2, 3\} \rightarrow S$
$\{a, b, c\}$	$1 \mapsto a, 2 \mapsto b, 3 \mapsto c$
$\{\square, \triangle, \circ\}$	$1 \mapsto \square, 2 \mapsto \triangle, 3 \mapsto \circ$
$\{\text{red}, \text{green}, \text{blue}\}$	$1 \mapsto \text{red}, 2 \mapsto \text{green}, 3 \mapsto \text{blue}$
$\{\pi, e, \sqrt{2}\}$	$1 \mapsto \pi, 2 \mapsto e, 3 \mapsto \sqrt{2}$

Table 1.1

Each map is one-to-one (distinct inputs go to distinct outputs) and onto (every element of S is hit). So all four sets are in $\overline{\{1, 2, 3\}}$.

1.5 Remark — The Class is Huge

The equivalence class $\overline{\{1, 2, 3\}}$ is not just large but *uncountable*: there are at least as many **3**-element sets as there are unordered triples of real numbers, since each triple $\{x, y, z\}$ with $x, y, z \in \mathbb{R}$ distinct is itself a **3**-element set. So picking *four* representatives barely scratches the surface — one could pick a different four every day and never repeat.

A useful mental image: the equivalence class of $\{1, 2, 3\}$ collapses every **3**-element set, regardless of what its elements are, into a single point representing the cardinality “three.” This is exactly the point of Section 8.1’s setup — the equivalence relation $\equiv$ throws away every feature of a set *except* its size.

1.6 Remark — Why Not Include $\{1, 2, 3\}$ Itself?

Strictly, $\{1, 2, 3\}$ *is* in its own equivalence class — the relation $\equiv$ is reflexive, so $\{1, 2, 3\} \equiv \{1, 2, 3\}$ via the identity map. So $\{1, 2, 3\}$ would be a perfectly valid fifth answer. We avoided it above only because the problem asks for sets *in the class*, and naming the representative back at us feels like a cheat. But it is correct.

1.7 Remark — Connection to $\mathbb{N}_k$

Section 8.1 singles out the “canonical” k -element set $\mathbb{N}_k = \{1, 2, 3, \dots, k\}$ as a representative of the finite cardinality k . The equivalence class of $\{1, 2, 3\}$ is precisely the class of $\mathbb{N}_3$, and is exactly what the textbook will later call the cardinal number **3**:

$$\mathbf{3} = \overline{\mathbb{N}_3} = \{S : S \equiv \{1, 2, 3\}\}.$$

Under this view, naming four sets in the class is the same as naming four sets of cardinality **3** — which is what we did.